

Question 1 - Definitions

The transition probabilities for rolling and holding:

$$P_{\{i,j,k,\text{flip}\}} = 1/6 (1 - P_{\{j,i,0\}} + P_{\{i,j,k+2\}} + P_{\{i,j,k+3\}} + P_{\{i,j,k+4\}} + P_{\{i,j,k+5\}} + P_{\{i,j,k+6\}})$$

$$P_{\{i,j,k,\text{hold}\}} = 1 - P_{\{j,i+k,0\}}$$

These equations reflect the probabilistic outcomes of either choosing to roll or hold at a given game state, where i is the current player's score, j is the opponent's score, and k is the current turn total. The roll expression accounts for the possibility of losing the turn (rolling a 1), or adding the result to the turn total, while the hold expression models the probability of passing the turn and adding the accumulated turn total to the player's score.

Question 2a - Evaluating $P_{\{99,99,0\}}$

Using the definition from Question 1 (and plugging in 1 for all terminal states) we get:

$$P_{98,99,0} = (1/6)(1 - P_{99,98,0}) + 5/6$$

Before solving algebraically, we will simplify our expression by ruling out one of the terms in the max function.

Let $x = 1 - P_{99,98,0}$. Since $P_{99,98,0} \in [0, 1]$, we know that $x \in [0, 1]$.

Now we compare the two arguments of the max function:

First term: x

Second term: $(1/6)x + 5/6$

We determine which is larger:

$$(1/6)x + 5/6 \geq x$$

$$\Leftrightarrow (1/6)x + 5/6 - x \geq 0$$

$$\Leftrightarrow (-5/6)x + 5/6 \geq 0$$

$$\Leftrightarrow 5/6 \geq (5/6)x$$

$$\Leftrightarrow 1 \geq x$$

This is always true since $x \leq 1$, so:

$$(1/6)(1 - P_{99,98,0}) + 5/6 \geq 1 - P_{99,98,0}$$

with equality only when $P_{99,98,0} = 0$.

Thus, the maximum is always achieved by the second term, and we simplify:

$$P_{98,99,0} = (1/6)(1 - P_{99,98,0}) + 5/6$$

$$P_{99,99,0} = \frac{1}{6}(1 - P_{99,99,0}) + \frac{5}{6}$$

$$\underset{-\frac{5}{6}}{X} = \frac{1}{6}(1 - \underset{-\frac{5}{6}}{X}) + \frac{5}{6}$$

$$\underset{-X}{X - \frac{5}{6}} = \frac{1}{6}(1 - \underset{-X}{X})$$

$$-\frac{5}{6} = \frac{1}{6}(1 - X) - X$$

$$-\frac{5}{6} = \frac{1}{6} - \frac{1}{6}X - X$$

$$-\frac{5}{6} = \frac{1}{6} - \frac{7}{6}X$$

$$\underset{-\frac{1}{6}}{-\frac{5}{6}} \quad \underset{-\frac{1}{6}}{-\frac{1}{6}}$$

$$\underset{-\frac{7}{6}}{-1} = \underset{-\frac{7}{6}}{-\frac{7}{6}X}$$

$$X = \frac{6}{7}$$

$$P_{99,99,0} = \frac{6}{7}$$

Question 2b - Evaluating $P_{\{98,99,0\}}$ and $P_{\{99,98,0\}}$

Using the definition from Question 1 (and plugging in 1 for all terminal states) we get:

$$P_{98,99,0} = (1/6)(1 - P_{99,98,00}) + 5/6$$

$$P_{99,98,0} = (1/6)(1 - P_{98,99,0}) + 5/6$$

Before solving algebraically, we will simplify our expressions by ruling out one of the terms in the max functions (for both expressions).

Let $x = 1 - P_{98,99,0}$. Again, since $P_{98,99,0} \in [0, 1]$, we know $x \in [0, 1]$.
(The same follows if we let $x = 1 - P_{99,98,0}$)

Compare the two arguments:

First term: x

Second term: $(1/6)x + 5/6$

We determine the larger term (Just as we did in part a):

$$(1/6)x + 5/6 \geq x$$

$$\Leftrightarrow (1/6)x + 5/6 - x \geq 0$$

$$\Leftrightarrow (-5/6)x + 5/6 \geq 0$$

$$\Leftrightarrow 5/6 \geq (5/6)x$$

$$\Leftrightarrow 1 \geq x$$

Which is always true. So again, the second term dominates the first (for both).

(except when $P_{98,99,0} = 0$, which cannot happen do to the constant term in our expressions: $5/6$)

Hence:

$$P_{98,99,0} = (1/6)(1 - P_{99,98,00}) + 5/6$$

and

$$P_{99,98,0} = (1/6)(1 - P_{98,99,0}) + 5/6$$

$$p_{98,99,0} = \frac{1}{6} (1 - p_{99,98,0}) + \frac{5}{6}$$

$$p_{99,98,0} = \frac{1}{6} (1 - p_{98,99,0}) + \frac{5}{6}$$

$$X = \frac{1}{6} (1 - y) + \frac{5}{6}$$

$$y = \frac{1}{6} (1 - x) + \frac{5}{6}$$

$$X = \frac{1}{6} \left(1 - \left(\frac{1}{6}X + \frac{5}{6} \right) \right) + \frac{5}{6}$$

$$X = \frac{1}{6} \left(1 - \frac{1}{6}X - \frac{5}{6} \right) + \frac{5}{6}$$

$$X = \frac{1}{6} - \frac{1}{36}X - \frac{5}{36} + \frac{5}{6}$$

$$X = 1 - \frac{5}{36} - \frac{1}{36}X$$

$$\frac{37}{36}X = \frac{31}{36}$$

$$X = \frac{1116}{1332}$$

$$X = \frac{31}{37}$$

$$y = \frac{1}{6} \left(1 - \frac{6}{37} \right) + \frac{5}{6}$$

$$y = \frac{1}{6} - \frac{6}{222} + \frac{5}{6}$$

$$y = 1 - \frac{1}{37}$$

$$y = \frac{36}{37}$$

$$p_{98,99,0} = \frac{31}{37}$$

$$p_{99,98,0} = \frac{36}{37}$$

Question 2c - Evaluating $P_{\{97,99,0\}}$, $P_{\{97,99,2\}}$, $P_{\{99,97,0\}}$

Using the definition from Question 1 (and plugging in 1 for all terminal states) we get:

1. $P_{97,99,0} = \max(1 - P_{99,97,0}, (1/6)(1 - P_{99,97,0}) + (1/6)(P_{97,99,2}) + 4/6)$
2. $P_{97,99,2} = \max(1 - P_{99,97,0}, (1/6)(1 - P_{99,97,0}) + 5/6)$
3. $P_{99,97,0} = \max(1 - P_{97,99,0}, (1/6)(1 - P_{97,99,0}) + 5/6)$

Before solving algebraically, we will simplify each of the expressions by ruling out one of the terms in the max function.

1. $P_{97,99,0} = \max(1 - P_{99,97,0}, (1/6)(1 - P_{99,97,0}) + (1/6)(P_{97,99,2}) + 4/6)$

We are comparing:

First term: $1 - P_{99,97,0}$

Second term: $(1/6)(1 - P_{99,97,0}) + (1/6)(P_{97,99,2}) + 4/6$

Let's assume for contradiction that the first term is the larger one.

Then:

$$1 - P_{99,97,0} > (1/6)(1 - P_{99,97,0}) + (1/6)(P_{97,99,2}) + 4/6$$

Multiply both sides by 6:

$$6(1 - P_{99,97,0}) > (1 - P_{99,97,0}) + P_{97,99,2} + 4$$

Simplify:

$$5(1 - P_{99,97,0}) > P_{97,99,2} + 4$$

But we know that $P_{97,99,2} \leq 1$, so the right-hand side is at most 5. Then:

$$\begin{aligned} 5(1 - P_{99,97,0}) &> 5 \\ \Rightarrow 1 - P_{99,97,0} &> 1 \\ \Rightarrow P_{99,97,0} &< 0 \end{aligned}$$

This is a contradiction, because $P_{99,97,0} \geq 0$.

So our assumption must be false, and the second term is always greater than or equal to the first. Hence, we simplify:

$$P_{97,99,0} = (1/6)(1 - P_{99,97,0}) + (1/6)(P_{97,99,2}) + 4/6$$

$$2. \quad P_{97,99,2} = \max(1 - P_{99,97,0}, (1/6)(1 - P_{99,97,0}) + 5/6)$$

Let $x = 1 - P_{99,97,0}$, with $x \in [0, 1]$.

Compare:

First term: x

Second term: $(1/6)x + 5/6$

As before:

$$(1/6)x + 5/6 \geq x$$

$$\Leftrightarrow (-5/6)x + 5/6 \geq 0$$

$$\Leftrightarrow 1 \geq x$$

So the second term is greater unless $P_{99,97,0} = 0$, in which case both are equal.

Therefore:

$$P_{97,99,2} = (1/6)(1 - P_{99,97,0}) + 5/6$$

$$3. \quad P_{99,97,0} = \max(1 - P_{97,99,0}, (1/6)(1 - P_{97,99,0}) + 5/6)$$

Let $x = 1 - P_{97,99,0}$, with $x \in [0, 1]$.

Compare:

First term: x

Second term: $(1/6)x + 5/6$

Again:

$$(1/6)x + 5/6 \geq x$$

$$\Leftrightarrow (-5/6)x + 5/6 \geq 0$$

$$\Leftrightarrow 1 \geq x$$

So the maximum is achieved by the second term.

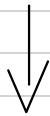
Thus:

$$P_{99,97,0} = (1/6)(1 - P_{97,99,0}) + 5/6$$

$$P_{97,99,0} = \frac{1}{6} (1 - P_{99,97,0}) + \frac{1}{6} P_{97,99,2} + \frac{4}{6}$$

$$P_{97,99,2} = \frac{1}{6} (1 - P_{99,97,0}) + \frac{5}{6}$$

$$P_{99,97,0} = \frac{1}{6} (1 - P_{97,99,0}) + \frac{5}{6}$$



$$X = \frac{1}{6}(1-Z) + \frac{1}{6}Y + \frac{4}{6}$$

$$Y = \frac{1}{6}(1-Z) + \frac{5}{6}$$

$$Z = \frac{1}{6}(1-X) + \frac{5}{6}$$

$$X = \frac{1}{6}(1-z) + \frac{1}{6}y + \frac{4}{6}$$

$$X = \frac{1}{6} \left(1 - \left(\frac{1}{6}(1-x) + \frac{5}{6} \right) \right) + \frac{1}{6} \left(\frac{1}{6} \left(1 - \left(\frac{1}{6}(1-x) + \frac{5}{6} \right) \right) + \frac{5}{6} \right) + \frac{4}{6}$$

$$X = \frac{1}{6} \left(1 - \frac{1}{6} + \frac{1}{6}x - \frac{5}{6} \right) + \frac{1}{6} \left(\frac{1}{6} \left(1 - \frac{1}{6} + \frac{1}{6}x - \frac{5}{6} \right) + \frac{5}{6} \right) + \frac{4}{6}$$

$$X = \frac{1}{6} \left(\frac{1}{6}x \right) + \frac{1}{6} \left(\frac{1}{6} \left(\frac{1}{6}x \right) + \frac{5}{6} \right) + \frac{4}{6}$$

$$X = \frac{1}{6} \left(\frac{1}{6}x \right) + \frac{1}{6} \left(\frac{1}{36}x + \frac{5}{6} \right) + \frac{4}{6}$$

$$X = \frac{1}{36}x + \frac{1}{216}x + \frac{5}{36} + \frac{4}{6}$$

$$X = \frac{7}{216}x + \frac{29}{36}$$

$$+ \frac{7}{216}x + \frac{7}{216}x$$

$$\frac{223}{216}x = \frac{29}{36}$$

$$\frac{223}{216} \quad \frac{223}{216}$$

$$X = \frac{6264}{8028}$$

$$X = \frac{174}{223}$$

$$Z = \frac{1}{6}(1-x) + \frac{5}{6}$$

$$Z = \frac{1}{6} \left(1 - \frac{174}{223} \right) + \frac{5}{6}$$

$$Z = \frac{1}{6} \left(\frac{49}{223} \right) + \frac{5}{6}$$

$$Z = \frac{49}{1338} + \frac{5}{6}$$

$$Z = \frac{49}{1338} + \frac{1115}{1338}$$

$$Z = \frac{1164}{1338}$$

$$Z = \frac{194}{223}$$

$$y = \frac{1}{6}(1-z) + \frac{5}{6}$$

$$y = \frac{1}{6} \left(1 - \frac{194}{223} \right) + \frac{5}{6}$$

$$y = \frac{1}{6} \left(\frac{29}{223} \right) + \frac{5}{6}$$

$$y = \frac{29}{1338} + \frac{5}{6}$$

$$y = \frac{29}{1338} + \frac{1115}{1338}$$

$$y = \frac{1144}{1338}$$

$$y = \frac{572}{669}$$

$$P_{97, 99, 0} = \frac{174}{223}$$

$$P_{97, 99, 2} = \frac{572}{669}$$

$$P_{99, 97, 0} = \frac{194}{223}$$